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Professional Summary

Highly accomplished and results-oriented Principal Data Architect with 12+ years of experience in designing, developing, and deploying robust and scalable data solutions leveraging cutting-edge AI/ML technologies. Proven ability to lead and mentor teams, translate complex business requirements into technical specifications, and deliver impactful solutions that drive significant business value. Expertise in architecting and implementing large-scale data pipelines, developing and deploying machine learning models, and managing cloud infrastructure on AWS. Passionate about leveraging AI/ML to solve complex business problems and create innovative solutions.

Education

- **University of Oxford**, Oxford, UK
  - Doctor of Philosophy (PhD) in Computer Science, 2011
  - Dissertation: "Novel Applications of Deep Learning in Natural Language Processing"
  - Master of Science (MSc) in Computer Science, 2009
  - Bachelor of Science (BSc) in Mathematics and Computer Science, 2007

Technical Skills

- **Programming Languages:** Java, C++, Go, Python, R, SQL
- **Machine Learning Frameworks:** TensorFlow, PyTorch, scikit-learn, XGBoost, Keras
- **Cloud Computing:** AWS (EC2, S3, RDS, Lambda, EMR, SageMaker), Azure (basic familiarity)
- **Big Data Technologies:** Hadoop, Spark, Hive
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn
- **Version Control:** Git

Professional Experience

- **Aetherium Analytics**, London, UK
  - **Principal Data Architect** | 2018 – Present
    - Led the design and implementation of a real-time fraud detection system using deep learning, resulting in a 25% reduction in fraudulent transactions.
    - Architected and deployed a large-scale data pipeline processing terabytes of data daily using Spark and AWS.

        - Mentored and guided a team of 5 junior data engineers and data scientists.
        - Developed and implemented a robust data governance framework to ensure data quality and compliance.
        - Successfully migrated on-premise data infrastructure to AWS cloud, reducing operational costs by 15%.
- **Quantium**, London, UK
    - **Senior Data Scientist** | 2015 – 2018
        - Developed and deployed machine learning models for customer churn prediction, resulting in a 10% increase in customer retention.
        - Conducted A/B testing to optimize marketing campaigns, leading to a 5% increase in conversion rates.
        - Collaborated with cross-functional teams to define business requirements and translate them into technical specifications.
- **Innovatech Solutions**, London, UK
    - **Data Scientist** | 2011 – 2015
        - Developed and implemented machine learning models for various applications, including image recognition, natural language processing, and time series forecasting.
        - Conducted data analysis and visualization to identify trends and insights.
        - Contributed to the development of new data science methodologies and tools.

AI/ML Projects

- **Real-time Fraud Detection System (Aetherium Analytics):** Developed a deep learning-based system for detecting fraudulent transactions in real-time. Utilized a combination of LSTM networks and convolutional neural networks to analyze transaction data and identify suspicious patterns. Achieved 95% accuracy in detecting fraudulent transactions. Technologies used: TensorFlow, AWS SageMaker, Python.

- **Customer Churn Prediction Model (Quantium):** Built a machine learning model to predict customer churn using various features such as customer demographics, purchase history, and engagement metrics. Utilized XGBoost and achieved an AUC score of 0.92. Technologies used: Python, scikit-learn, XGBoost.

- **Image Recognition System (Innovatech Solutions):** Developed a convolutional neural network (CNN) for image recognition using TensorFlow. Achieved 90% accuracy on a benchmark dataset. Technologies used: TensorFlow, Python.

- **Natural Language Processing (NLP) System (PhD Dissertation):** Developed novel algorithms for sentiment analysis and text summarization using recurrent neural networks. Published research findings in leading academic journals. Technologies used: Python, TensorFlow, various NLP libraries.